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EXPERIMENT - 7 IPC(SHARED MEMORY)

AIM:

To write and execute a C program for Inter Process Communication using Shared Memory.

CODE:

```
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
#include<string.h>
int main(){
    key_t key=1234;
    int shmid;
    shmid=shmget(key,1024,0666|IPC_CREAT);
    char *str=(char*)shmat(shmid,(void*)0,0);
    printf("Enter message: ");
    scanf("%[^\n]",str);
    printf("Data written to shared memory: %s\n",str);
    return 0;
}
```

"sender.c" 14L, 341B

```
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
int main(){
    key_t key=1234;
    int shmid;
    shmid=shmget(key,1024,0666);
    char *str=(char*) shmat(shmid,(void*)0,0);
    printf("Data read from shared memory: %s\n",str);
    shmdt(str);
    return 0;
}
```

"receiver.c" 12L, 278B

OUTPUT:

A terminal window titled 'USER@123@localhost:~' with standard window controls (minimize, maximize, close) on the right. The terminal shows a sequence of commands and their outputs. The user enters 'vi sender.c', 'vi receiver.c', 'gcc sender.c', and './a.out'. After './a.out', the prompt 'Enter message: ' appears, and 'hello' is entered. The output 'Data written to shared memory: hello' is shown. Then, the user enters 'gcc receiver.c', './a.out', and the output 'Data read from shared memory: hello' is displayed. The prompt returns to '[USER@123@localhost ~]\$' with a green cursor.

```
[USER@123@localhost ~]$ vi sender.c
[USER@123@localhost ~]$ vi receiver.c
[USER@123@localhost ~]$ gcc sender.c
[USER@123@localhost ~]$ ./a.out
Enter message: hello
Data written to shared memory: hello
[USER@123@localhost ~]$ gcc receiver.c
[USER@123@localhost ~]$ ./a.out
Data read from shared memory: hello
[USER@123@localhost ~]$ █
```

RESULT:

Thus, the Inter Process Communication using Shared Memory was implemented successfully and the expected output was obtained.